

Ultra-wideband MHz to THz plasmonic EO modulator

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Electromagnetic fields at terahertz frequencies open a window to multiple applications, such as wireless communication, sensing, and medical imaging. Yet the challenge remains to efficiently generate, receive and process such high frequencies. In this work, we investigate plasmonic modulators to bring THz frequencies to photonic integrated circuits (PICs). We demonstrate a frequency response beyond the 1 THz and develop an equivalent circuit model of the modulator. Our results show that plasmonic modulators are uniquely suited for THz PICs.

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THz frequencies (0.1–10 THz) show high potential in applications ranging from wireless communication [1], biology and medical sciences (i.e., sensing, spectroscopy, and imaging) [2,3] to security [3]. Of particular interest is THz wireless communications, which could bridge the capacity gap between wireless and optical fiber networks, paving the way for 6G mobile networks [4]. The inherent high available bandwidth at THz carrier frequencies enables high data rates [5–9]. Furthermore, the high robustness toward atmospheric turbulence and visibility-limiting weather conditions, such as fog, snow, haze, or dust, already enabled to overcome free-space distances of up to 5 km [10] in urban scenarios. In THz communication research, there are three main approaches: the all-electrical-based [11,12], the hybrid electrical-photonic-based [5–9,13], and the all-photonic-based [8,9,14–16] THz generation and detection. The electrical-based approach enabled the first THz links by using sources (electrical multiplier chains) and sensitive detectors based on technologies, such as high-electron-mobility transistors (HEMTs) [11] and Schottky barrier diodes (SBDs) [12]. In the hybrid approach, one leverages high-speed photodetectors based on III–V [17] or graphene [18] at the transmitter (Tx) side to generate broadband THz signals [13], while on the receiver (Rx) side the electrical mixers are kept due to their high sensitivity (NEP of ~ 1.5 pW/ $\sqrt{\text{Hz}}$ [19]). Typically,

multiple THz channels were necessary as electrical THz receivers are limited by a few tens of GHz intermediate frequency (IF) bandwidth, non-flat frequency response [6], and non-linearities (N th harmonic mixing) [7].

The photonic approach resolves these issues on the receiver side by utilizing electro-optical (EO) modulators to directly and linearly convert the THz signal into the optical domain. This enables seamless integration into the high-capacity fiber optic backbone of the Internet, eliminating the demanding THz-to-IF conversion. Yet the challenge lies in achieving THz EO bandwidths. Conventional EO modulators (LiNbO₃, InGaAs, and Si) so far limited the wireless carrier frequencies to ~ 100 GHz and its signal bandwidths to 14 GHz [16].

Recent developments in plasmonics [20] have led to plasmonic modulators that offers a flat frequency response from the DC to up to 500 GHz and potentially beyond [21]. Furthermore, they feature high modulation efficiencies with $V_{\pi}L$ as low as 48 V μm [22]. In 2015, a novel direct THz-to-optical converter was introduced [23], where a THz field is received by an on-chip antenna and efficiently encoded onto an optical carrier by the seamlessly integrated plasmonic modulator. The flat ultra-high bandwidth of the plasmonic modulator allows operation at frequencies of >1 THz [24] and offers flexibility in choosing between on-chip resonant antennas [15] or high-directivity off-the-shelf antennas [9].

In the past, the high-speed behavior of plasmonic modulators has been exploited in fiber systems to achieve the highest data rates [25]. However, plasmonic modulators do not only constitute an interesting candidate for highest-speed fiber communications but also for new applications such as THz communications [8,9,15]. They offer a simple, linear, ultra-broadband (DC to THz) mixer solution within a single device. Furthermore, the high and frequency-independent modulation efficiency make plasmonic modulators attractive as a sensitive THz sensor in spectroscopy, imaging, or radar applications [26].

In this work, we present a broadband EO modulator operating from 10 MHz to 1.14 THz. This device features a 3 dB EO bandwidth of 997 GHz, which, to the best of our knowledge, is the largest bandwidth achieved by any EO device to date. This

was achieved through an optimized modulator design, expanding the tested EO frequency range to more than double that of previous measurements. Furthermore, we investigate the whole THz circuit (50 Ω -matched THz source, THz probe, and plasmonic modulator) by means of modeling and simulations, matching the measured response. From this, we root out the modulators current limitations and propose how to reach operation at frequencies above 1 THz by further design optimization. As a result, our findings show that plasmonic modulators offer an immense bandwidth beyond 1 THz and are uniquely suited for THz PICs.

Figure 1(a) shows the experimental setup that was used to assess the frequency response of the plasmonic modulator, fabricated on polariton's plasmonic PIC platform [27]. A tunable laser source provides a -6 dBm optical carrier in the C-band and is coupled to the PIC via a single-polarization (TE) grating coupler (GC). The light propagates through the Si waveguides to the plasmonic modulator. The optical field is highly confined in the plasmonic metal-insulator-metal (MIM) slot and interacts with the nonlinear organic electro-optic (OEO, light green) material filling the slot. A THz frequency signal (f_{THz}) is generated by an external source and fed to the plasmonic modulator where the THz field is also highly confined in the same region as the optical field. By this, the THz signal is encoded onto the optical carrier using the Pockels effect. The modulated optical signal is coupled into a Si waveguide and then out of the PIC via another GC. The modulated light is finally analyzed using an optical spectrum analyzer (OSA). Eight different sources based on electrical mixers were utilized to generate sinusoidal THz signals across the investigated frequency range from 10 MHz to 1.14 THz (see Supplement 1). The frequency response of the modulator below 10 MHz is flat [28]. In this experiment,

the lower frequency response measurement was limited by the resolution of the OSA.

The modulator consists of two parallel plasmonic phase shifters connected via GSG contact pads, directly applying the THz field across 10–15 μm long slots with 100 nm widths. The GSG configuration aligns with the design of the electrical probes and supports performance assumptions for future Mach–Zehnder modulators (MZM). The 10- μm -long phase modulator exhibits insertion losses of 5.6 dB and $V_{\pi, \text{PM}}$ of 234 $\text{V}\mu\text{m}$ (this corresponds to a 117 $\text{V}\mu\text{m}$ for a MZM). The fiber-to-chip coupling loss was 2.9 dB/coupler with a 3 dB optical bandwidth of ~ 45 nm (both grating couplers).

The OSA was used to measure the peak-to-sideband power ratio of the modulated signal and, thus, the modulation efficiency. The calibrated modulation efficiency η is plotted in Fig. 1(b) on a logarithmic scale from 10 MHz to 100 GHz, demonstrating the flat frequency response for typical optical communication scenarios and further extended in Fig. 1(c) from 10 MHz to 1.2 THz on a linear scale. We observe that Device 1 (red) and Device 2 (blue) feature measured 3 dB bandwidths of 880 GHz and 997 GHz, respectively. Device 1 has a $25 \mu\text{m} \times 25 \mu\text{m}$ signal pad size, compatible with electrical probes that have a pitch of 25–50 μm . Device 2 (blue curve) has a $20 \mu\text{m} \times 25 \mu\text{m}$ signal pad size, compatible with 25 μm pitch electrical probes [see the insets in Fig. 1(c)]. A peaking behavior between 500 GHz and 800 GHz is observed in both cases, likely due to parasitic inductances. To understand this and explore potential improvements, we derive a lumped-element model, justified by the fact that the frequency of 1 THz corresponds to an in-device wavelength of $\sim 100 \mu\text{m}$, much greater than the 10 μm slot length.

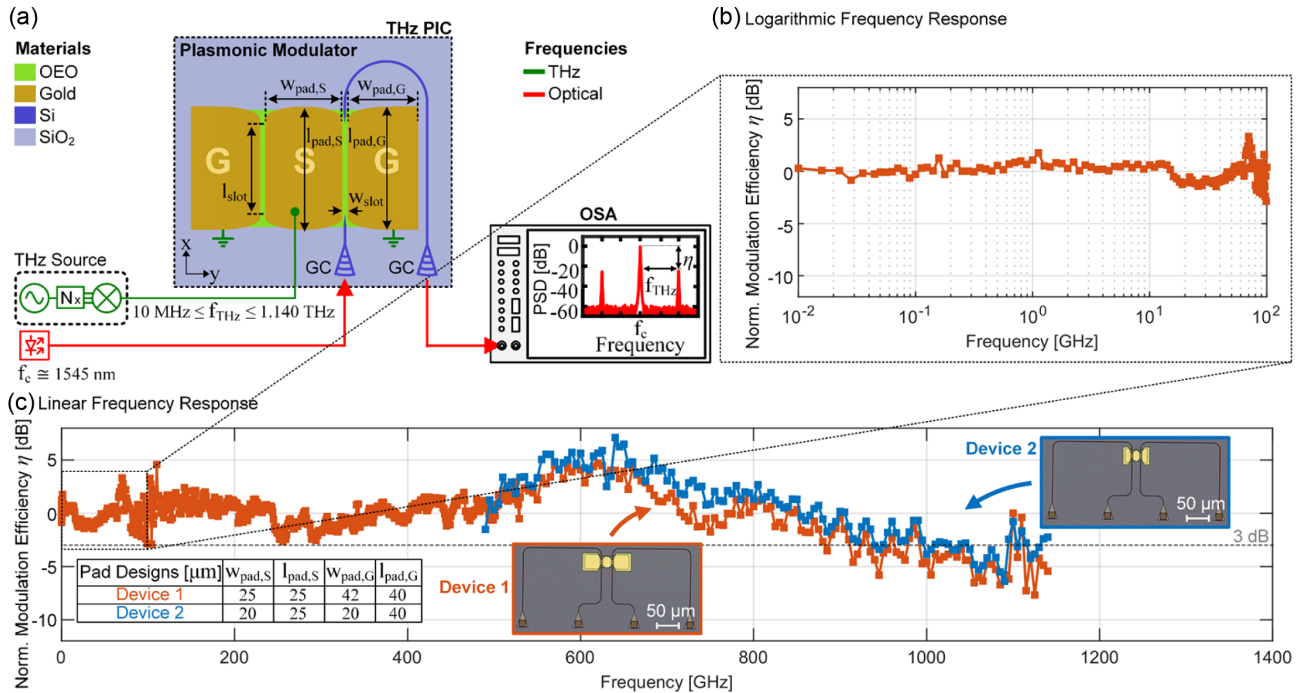


Fig. 1. THz modulation experiment. (a) EO plasmonic modulator in a THz bandwidth characterization setup. A sinusoidal signal at frequency $10 \text{ MHz} \leq f_{\text{THz}} \leq 1.14 \text{ THz}$ is generated with a THz source and fed through the plasmonic modulator. The signal is encoded onto an optical carrier $f_{\text{opt}} \approx 1550 \text{ nm}$ by the plasmonic modulator. The response is read out with an optical spectrum analyzer (OSA). Note that the plasmonic modulator is used as a phase shifter because the optical path only passes through one arm of the pads that are in a GSG configuration. (b) Zoom-in to the range from 10 MHz to 100 GHz. (c) Full range from 10 MHz to 1.2 THz for two device designs. Device 1 (red) with a signal pad size of $25 \mu\text{m} \times 25 \mu\text{m}$ compatible with 25 μm and 50 μm pitch electrical probes. Device 2 (blue) with a signal pad size of $20 \mu\text{m} \times 25 \mu\text{m}$ compatible with 25 μm pitch electrical probes.

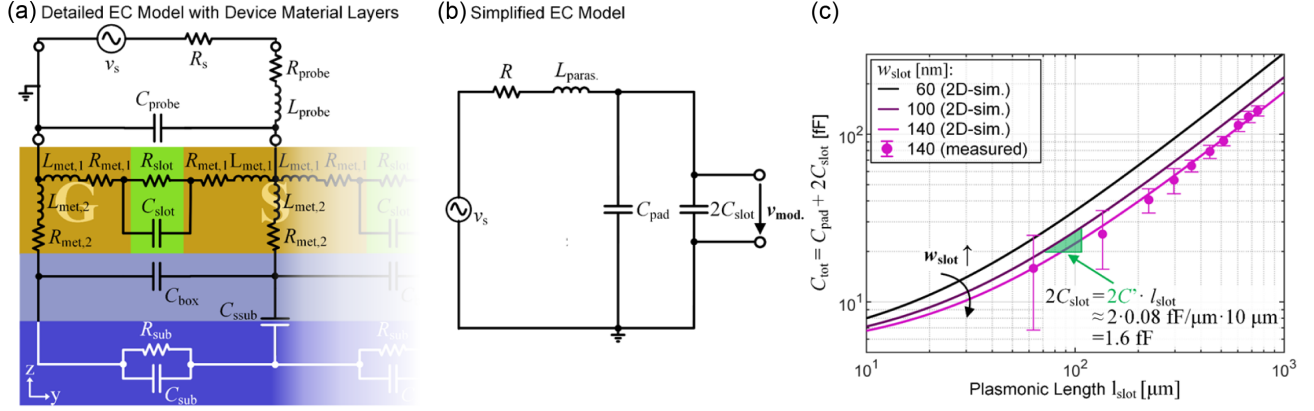


Fig. 2. Modeling of the plasmonic modulator. (a) Detailed equivalent circuit (EC) respecting all material layers and (b) simplified model of the plasmonic phase shifters in the GSG configuration. (c) The capacitance $C_{\text{tot}} = C_{\text{pad}} + 2C_{\text{slot}}$ of the plasmonic MZM as a function of the plasmonic length l_{slot} . For this, reference devices ($w_{\text{slot}} = 140$ nm and signal pad $42 \mu\text{m} \times 50 \mu\text{m}$) were fabricated, and low-frequency (1 MHz) capacitance measurement were performed. We observed a good agreement of the measurements with the 2D simulation and derived a C_{slot} of ~ 1.6 fF for our plasmonic modulators in the THz Exp.

The complete equivalent circuit (EC) model of the plasmonic phase shifters in ground-signal-ground (GSG) configuration is depicted in Fig. 2(a), following [29]. The second plasmonic phase shifter is not shown due to symmetry. Assuming that resistances R_{sub} ($\sim \text{k}\Omega$) and R_{slot} ($\sim \text{M}\Omega$) are dominated by the parallel capacitances C_{sub} and C_{slot} , respectively, we can derive a simplified EC model for a MZM, as shown in Fig. 2(b). Note that $L_{\text{met},1}$ and $L_{\text{met},2}$ are represented here now by the parasitic inductance $L_{\text{paras.}}$. Furthermore, $R_{\text{met},1}$, $R_{\text{met},2}$, and R_s (typ. 50Ω) are summarized by R . This way, we can derive the voltage transfer function $t(\omega)$ for the simplified circuit in Fig. 2(b),

$$t(\omega) = \frac{v_{\text{mod.}}(\omega)}{v_s(\omega)} = \frac{1}{1 + j\omega \cdot (C_{\text{pad}} + 2C_{\text{slot}}) \cdot (R + j\omega L_{\text{paras.}})} \quad (1)$$

Thus, the voltage over the MZM describes an RLC resonance, whose 3-dB cutoff frequency f_{cut} can be approximated by

$$f_{\text{cut, est.}} \approx \frac{1}{2\pi R \cdot (C_{\text{pad}} + 2C_{\text{slot}})} \quad (2)$$

Finding the slot capacitance C_{slot} and pad capacitance C_{pad} therefore allows us to model the frequency limitation.

To obtain the C_{slot} , devices with varying slot lengths but fixed pad layouts (signal pad: $42 \mu\text{m} \times 50 \mu\text{m}$) and fixed slot widths of 140 nm were fabricated and characterized. The capacitance was measured using a multi-frequency capacitance measurement unit. The total measured capacitance C_{tot} , including contributions from the pad and the slot, is plotted as filled circles in Fig. 2(c). With the measured data, we modeled the plasmonic slot in a 2D-electrostatic simulation. This allows for the calculation of the C_{tot} for slot widths other than 140 nm. The simulations are shown as the solid lines in Fig. 2(c). The simulated and measured data overlap well. Next, we assume that the slot length only influences the slot capacitance C_{slot} and that this influence is linear, i.e.,

$$\frac{d}{dl_{\text{slot}}} C_{\text{tot}} = 2C' = \frac{2C_{\text{slot}}}{l_{\text{slot}}} \quad (3)$$

Simulations with other slot widths (60 nm and 100 nm) then allow the extraction of the capacitance per unit length C' of the respective slot width, assuming the linear relationship

$2C_{\text{slot}} = 2C' l_{\text{slot}}$. From this, we derive the capacitance of the measured THz modulator. The $2C_{\text{slot}}$ is 1.6 fF for a 100 nm slot width and 10 μm length, which aligns well with the previously reported value of ~ 3.4 fF for MZMs with 130 nm wide, 20 μm long slots [30].

The pad design significantly impacts the modulator's overall response since C_{pad} is typically larger or similar to the device capacitance $2C_{\text{slot}}$. The influence of the pad design on the frequency response is studied using 3D FEM electrical simulations, with results summarized in Figs. 3(a) and 3(b).

Figure 3(a) shows the electrical response of the pads combined with the plasmonic device for the varying pad designs and slot widths in the frequency range from 0.01 to 1.4 THz. The plots with the red and blue lines correspond to the pad dimensions of Devices 1 and 2 in Fig. 1. The simulations align very well with the measurement results. In addition, we see that smaller pads yield significantly higher bandwidths due to the reduced C_{pad} . Furthermore, smaller pad designs increase the influence of C_{slot} and thus the w_{slot} on the bandwidth, as Eq. (2) indicates. The fit over the 3D FEM results for $w_{\text{slot}} = 100$ nm in Fig. 3(b) shows the required pad size dimensions to reach specific cutoff frequencies. The peaking behavior in Fig. 3(a) emerges again and confirms the origin from a parasitic inductance. It increases the cutoff frequency to some extent. To further explore this, we simulate the simplified equivalent circuit, see Fig. 2(c), and study the effects of $L_{\text{paras.}}$ and R for varying C_{pad} on the cutoff frequency f_{cut} . The results are plotted in Fig. 3(c) and 3(d), respectively. We find the best match with both, our THz measurements and our 3D simulations for $L_{\text{paras.}} = 15$ pH, resulting in a C_{pad} of ~ 2.2 fF and 1.8 fF for Device 1 and Device 2, respectively.

In Fig. 3(c), we observe that while decreasing R , the achievable cutoff frequencies f_{cut} increase as well. As expected, the increase does not fully align with the predictions of our Eq. (2) due to the influence of the parasitic inductance $L_{\text{paras.}}$. From Eq. (1), however, we see that the term $\omega L_{\text{paras.}}$ can dominate the resistance R . This means that for a fixed $L_{\text{paras.}}$, the cutoff frequency f_{cut} does not increase arbitrarily, by further reducing the resistance R .

Next, we simulated our equivalent circuit for a varying parasitic inductance $L_{\text{paras.}}$, as shown in Fig. 3(d). We found that Eq. (2) only holds well for all $L_{\text{paras.}}$ if $C_{\text{pad}} \geq 2$ fF. Additionally, leveraging the

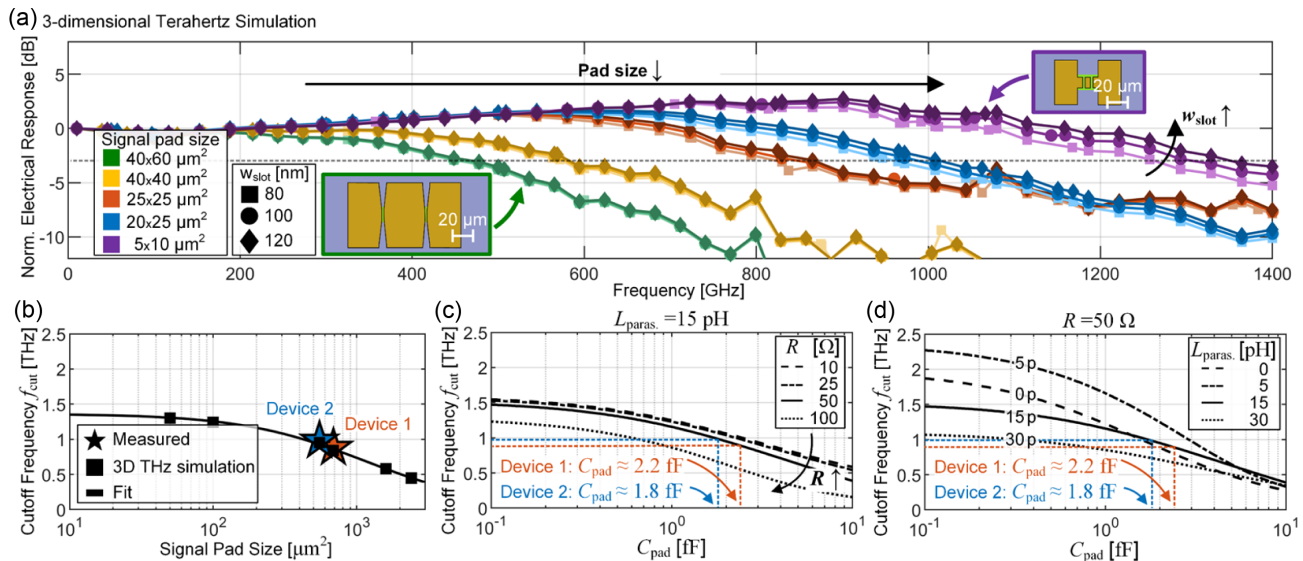


Fig. 3. Electrical 3D simulation of the device, investigating the influence of the pad capacitance C_{pad} . (a) Simulated normalized frequency response of the plasmonic modulators with different signal pad sizes (green, red, blue, and purple) for different slot widths w_{slot} , all designed for a 25 μm pitch GSG electrical probe. (b) Cutoff frequency as a function of the signal pad size. The solid curve corresponds to a model derived from Eq. (2). (c) Simulations of the simplified EC in Fig. 2(b). The resulting cutoff frequencies are plotted as a function of the C_{pad} for varying (c) L_{paras} and varying (d) R .

LC resonance effect, a small L_{paras} (e.g., 5 pH) can boost the cutoff frequency f_{cut} by $\sim 40\%$, reaching up to 1.4 THz.

In our current experiments, practical limitations, such as electrical probe contacting, restrict bandwidths to 1 THz. However, by $C_{\text{pad}} \rightarrow 0$, thus removing the pads through, e.g., a monolithic device design [31], the f_{cut} could reach up to 1.9 THz for $R \approx 50$ Ω. Alternatively, transitioning from a rigid THz waveguide-based setup to a wireless scheme, where the THz field is detected by the on-chip antenna-integrated plasmonic modulator (a quasi RLC circuit) [24,26], would allow for optimal tuning of R and L_{paras} to make operation frequencies of up to 2.4 THz [24] accessible.

This work demonstrates the first broadband EO modulation from 10 MHz to 1.14 THz using an EO plasmonic modulator, achieving a record 3 dB bandwidth of 997 GHz. This was made possible by an optimized modulator design, doubling the tested EO frequency range compared to previous efforts. This study also identifies current limitations and proposes solutions for operation above 1 THz, highlighting the potential of plasmonic modulators for terahertz PICs.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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